

- 3 (a) (i) The gravitational force of attraction that Earth exerts on the satellite provides the centripetal force required for the satellite's circular motion about the Earth.

$$\frac{GMm}{R^2} = \frac{mv^2}{R}$$

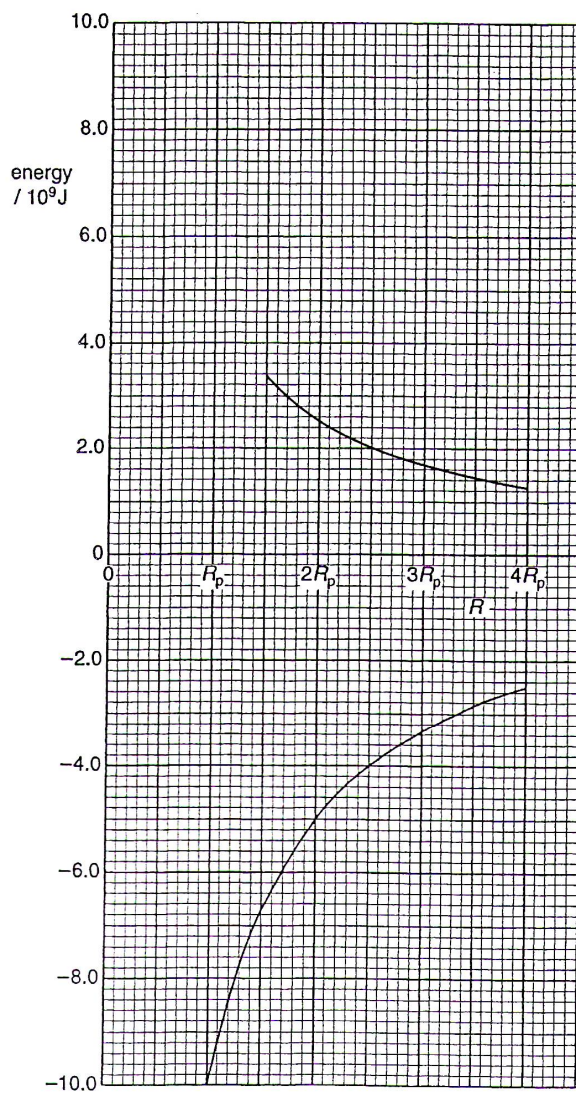
$$\frac{GMm}{R} = mv^2$$

$$\text{Kinetic energy } E_K = \frac{1}{2}mv^2 = \frac{GMm}{2R}$$

(ii) Gravitational potential energy $E_P = -\frac{GMm}{R}$

$$\text{Therefore } \frac{E_P}{E_K} = -2 \text{ (shown)}$$

(b) (i)



(ii) When $R = 4R_p$, $E_p = -2.5 \times 10^9$ J.

Since $\frac{E_p}{E_k} = -2$, $E_k = 1.25 \times 10^9$ J.

$$\frac{1}{2}(1600)v^2 = 1.25 \times 10^9$$

$$v = 1250 \text{ ms}^{-1}$$

When $R = 2R_p$, $E_p = -5.0 \times 10^9$ J.

Since $\frac{E_p}{E_k} = -2$, $E_k = 2.5 \times 10^9$ J.

$$\frac{1}{2}(1600)v^2 = 2.5 \times 10^9$$

$$v \approx 1770 \text{ ms}^{-1}$$

The change in speed $= 1770 - 1250 \approx 520 \text{ ms}^{-1}$

Note: Do not equate the change in kinetic energy with the change in potential energy.